

LAB SAFETY

SCENARIO ANALYSIS

MAKE THE SAFE CALL

WHAT STUDENTS DO

Analyze 10 original laboratory situations, identify evidence of risk, choose the safest first action, and justify decisions using a consistent safety framework.

GRADES 9–11	CORE 45–55 MIN	PRINT + EDITABLE
CORE + HONORS	EXIT TICKET	FULL KEY

Original content • Printer-friendly • Structured reasoning

Scenarios 1-4

Notice the evidence, state the risk, choose the safest first action, and explain why it works.

1. Goggles on Standby

Maya begins measuring a dilute solution while her splash goggles rest on her forehead. Her partner is still collecting materials.

Evidence noticed: _____

Risk / possible harm: _____

Safest first action: _____

Why this action reduces risk: _____

2. Blocked Aisle

Two backpacks and a charging cable cross the aisle between a lab table and the exit.

Evidence noticed: _____

Risk / possible harm: _____

Safest first action: _____

Why this action reduces risk: _____

3. Mystery Beaker

An unlabeled beaker containing a clear liquid is left beside the balance. No one in the group knows who placed it there.

Evidence noticed: _____

Risk / possible harm: _____

Safest first action: _____

Why this action reduces risk: _____

4. Heating Direction

Eli warms a test tube while its opening points toward the student across the table.

Evidence noticed: _____

Risk / possible harm: _____

Safest first action: _____

Why this action reduces risk: _____

Scenarios 1-4

Reasonable equivalent wording is acceptable when it follows local policy and does not add risk.

1. Goggles on Standby

Evidence: Chemical measuring has begun while Maya's goggles are on her forehead.

Risk: A splash could reach unprotected eyes.

Action: Stop measuring and wear splash goggles correctly before continuing.

Why: The eye barrier must be in place before exposure can occur.

2. Blocked Aisle

Evidence: Backpacks and a cable cross a travel route leading toward the exit.

Risk: Someone could trip, and movement during an emergency could be delayed.

Action: Stop traffic through the area and move the items to the assigned storage location.

Why: Removing the obstruction eliminates the trip hazard and keeps the exit route accessible.

3. Mystery Beaker

Evidence: The beaker has no label, and no one can identify the clear liquid.

Risk: Touching or testing an unknown substance could cause exposure or contamination.

Action: Do not touch or test it; keep others away and notify the teacher immediately.

Why: Avoiding contact prevents additional exposure while the teacher secures and identifies the material by site procedure.

4. Heating Direction

Evidence: The opening of a heated test tube points toward another person.

Risk: Hot liquid or vapor could eject toward that person.

Action: Stop heating; use the teacher-approved holder and point the opening away from all people.

Why: Stopping removes the immediate energy input, and safe orientation reduces direct exposure if material ejects.